

Maternal Health Data Analytics

Executive case study • Turning patient-level programme data into actionable insights

13,088 Patients analyzed	4 Districts	3 Programme phases	15% Patients under 20
33% Highest district anaemia rate	~40% High-risk women referred	49% → 77% HIV testing improvement	83% → 97% Hb testing improvement

Executive insight

The central finding is a risk-to-action gap: the analysis identified a substantial population of high-risk women, yet only about 40% were referred. This shifts the analytical question from “Who is at risk?” to “Where is the care pathway breaking down?”

Key findings

Demographics	15% of patients were under 20, highlighting the importance of adolescent maternal-health interventions.
Nutrition & anaemia	23% were underweight; Barpeta had the highest observed anaemia rate at 33%.
Screening gaps	Coverage ranged from 97.5% for BP checks to just 1.3% for TB screening.
Programme change	HIV testing improved from 49% to 77% and Hb testing from 83% to 97% between Baseline and Implementation.
Geographic variation	Districts showed distinct teen-pregnancy, anaemia and high-risk profiles, suggesting interventions should be locally targeted.

Screening coverage: where attention is needed

Service / indicator	Coverage
BP check	97.5%
Hb test	90.0%
HIV testing	49% → 77%
TB screening	1.3%
Malaria screening	2.6%
IFA supplementation	8.9%

Data quality snapshot

Data-quality check	Finding
Invalid ANC/LMP date pairs	31
Missing BMI inputs	403
Literacy / education	~36% Unknown

Analytical approach

Excel was used for structured analysis, cross-tabulation, calculations and visualization. The results were independently checked with Python/Pandas to validate the headline figures and reduce the risk of spreadsheet formula or calculation errors.

5 takeaways 1. Maternal risk varies meaningfully by district. 2. Screening coverage is strong for some core services but very weak for others. 3. Programme implementation is associated with substantial gains in HIV and Hb testing. 4. The referral pathway deserves attention: identification does not consistently translate into action. 5. Data-quality checks are essential before using programme data for decisions.
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Portfolio case study based on the accompanying workbook. Headline figures reflect the validated analysis provided for this case study.